Title - Interpreting Black-Box Models: A Practical Guide to SHAP with XGBoost

1. Introduction

Machine learning models like gradient boosting algorithms often demonstrate high predictive performance but are often treated as black boxes This opacity impedes their uptake in crucial fields like healthcare, finance, and decision-making systems domains in which it is almost as critical to know why a prediction was made as it is to receive the prediction itself.

This tutorial demonstrates how to enhance the interpretability of your model using SHAP (SHapley Additive Explanations) on XGBoost classifier. The objective is to display how SHAP can yield global insight (feature importance over the entire dataset) and local insight (prediction at the individual level).

SHAP is based on cooperative game theory, which provides consistent explanations that are well-founded on theoretical grounds, unlike traditional feature importance techniques. This tutorial discusses the use of SHAP to unveil the feature interactions of prediction methods in Machine Learning.

2. Background

2.1 XGBoost:

XGBoost is an ensemble learning algorithm based on gradient boosting decision trees. New trees are built sequentially to fix errors made in the previous ones. XGBoost is efficient, scalable and has great predictive power hence it is widely used.

This complexity makes direct interpretations of the model difficult, especially for features that are not linear across a store.

2.2 Model Interpretability:

Model interpretability refers to the ability to understand how input features influence model predictions. Traditional approaches, such as feature importance scores, provide limited insight because they do not explain direction (positive/negative impact) or interaction effects.

2.3 SHAP (SHapley Additive Explanations):

SHAP is based on Shapley values from cooperative game theory. It assigns each feature a contribution value for a given prediction.

Key advantages of SHAP:

- Provides consistent and fair feature attribution.
- Explains both global and local behaviour.
- Captures feature interactions.

3. Methodology

→3.1 Dataset:

This data analysis employs the breast cancer Wisconsin dataset to predict the classification of a tumour as malignant or benign.

- Features: 30 numerical attributes (e.g., mean radius, texture)
- Target: Binary classification (malignant vs benign)

The dataset is suitable for interpretability analysis due to its structured features and real-world relevance in healthcare.

3.2 Data Preprocessing

The dataset was so clean that no preprocessing was required at all. The following steps were undertaken to guarantee robustness and reproducibility.

- **Feature Representation:** The features were converted into Pandas Data Frame with appropriate column names for easy interpretation.

- **Target Encoding:** The target variable was kept as a binary numeric variable appropriate for classification.

- **Train-Test Split:** The splitting of the dataset into training (80%) and testing (20%) datasets was performed using stratified sampling to ensure that the ratio of malignant and benign cases is consistent throughout the two datasets.

- **Random State Fixing:** A random_state of 42 ensured that results were reproducible for any run of the code.

No feature scaling was applied, as tree-based models such as XGBoost are invariant to monotonic transformations of input features.

3.3 Model Selection and Training:

The model used in this study is **XGBoost**, a state-of-the-art ensemble learning method based on gradient boosting of decision trees.

Model Configuration:

The following hyperparameters were selected:

- n_estimators = 150 → Controls number of trees
- max_depth = 4 → Prevents overfitting by limiting tree complexity
- learning_rate = 0.05 → Ensures gradual learning
- subsample = 0.8 → Adds randomness for generalisation
- colsample_bytree = 0.8 → Reduces feature over-reliance

The parameters were selected to achieve a balance between the performance and interpretability of the model.

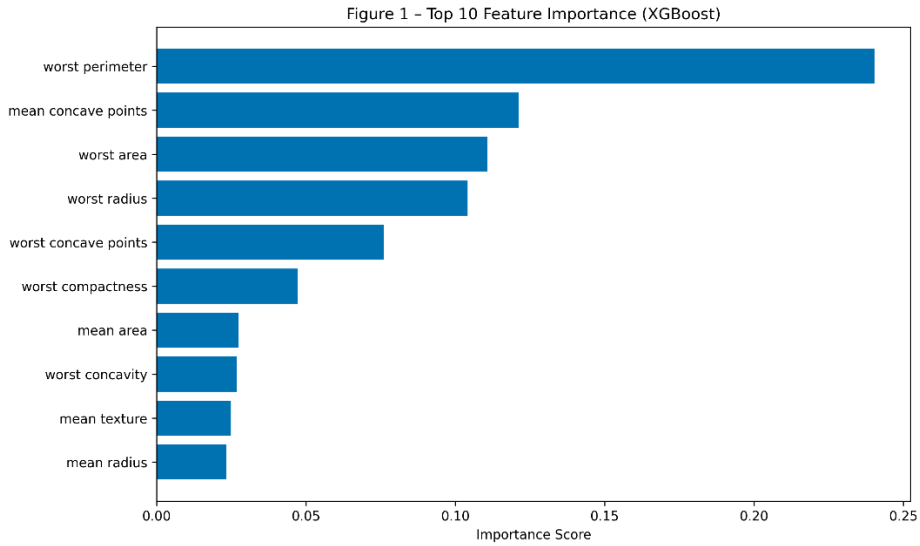
3.4 Model Evaluation Strategy:

The trained model was evaluated using multiple performance metrics:

- **Accuracy**: Measures overall classification correctness
- **ROC-AUC Score**: Evaluates model's ability to distinguish between classes.
- **Classification Report:** Includes precision, recall, and F1-score.

4. Results and Visualisations

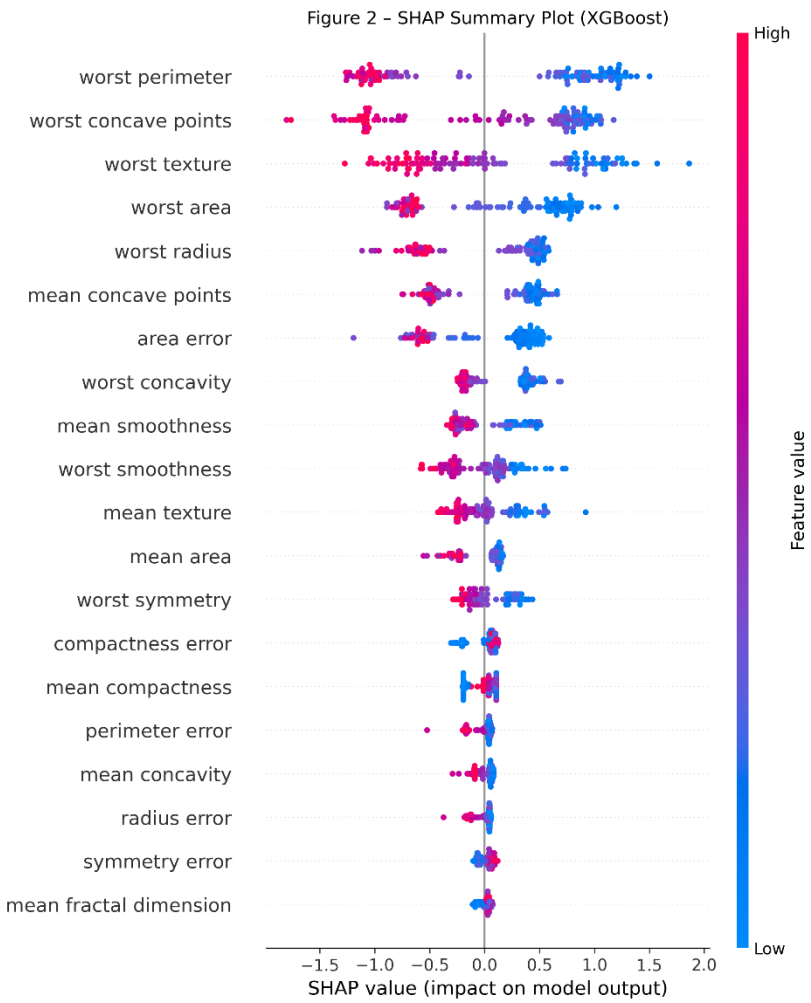
Figure 1 – Feature Importance:



This plot shows the most influential features identified by the XGBoost model. Features such as mean radius, mean texture, and mean perimeter have the highest importance scores, indicating their strong contribution to the classification task.

While useful for identifying key features, this method only provides a ranking. It does not explain the direction of influence (positive or negative impact) or how feature effects vary across different observations.

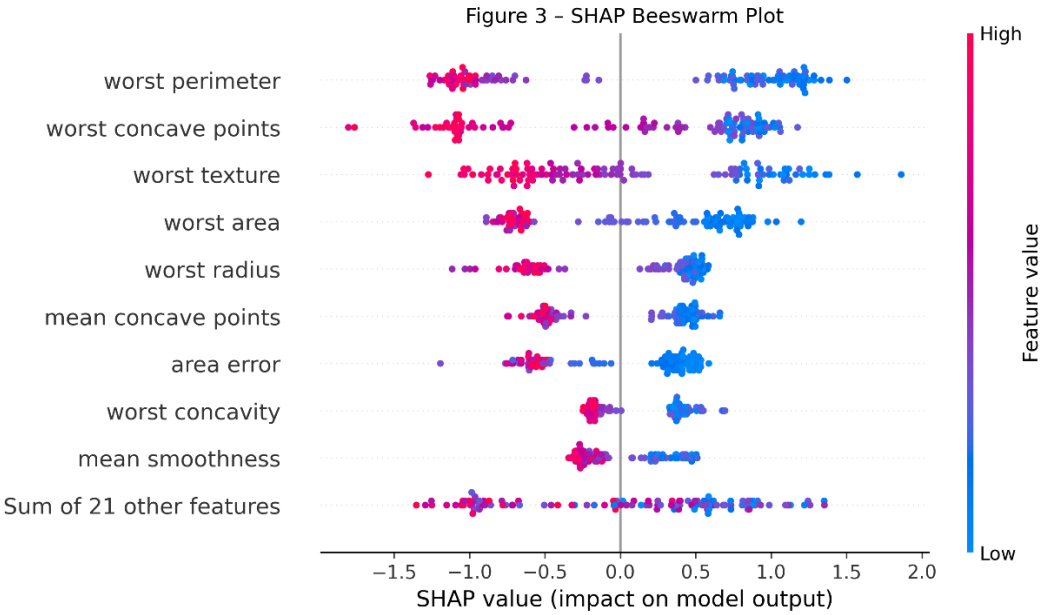
Figure 2 – SHAP Summary Plot:



The SHAP summary plot provides a comprehensive overview of feature impact across all samples. Each point represents an individual observation, where the horizontal position indicates the contribution to the prediction and the colour represents the feature value.

A clear pattern can be observed where higher values of features such as mean radius push predictions toward the malignant class. This plot is more informative than traditional feature importance as it shows both the magnitude and direction of feature influence.

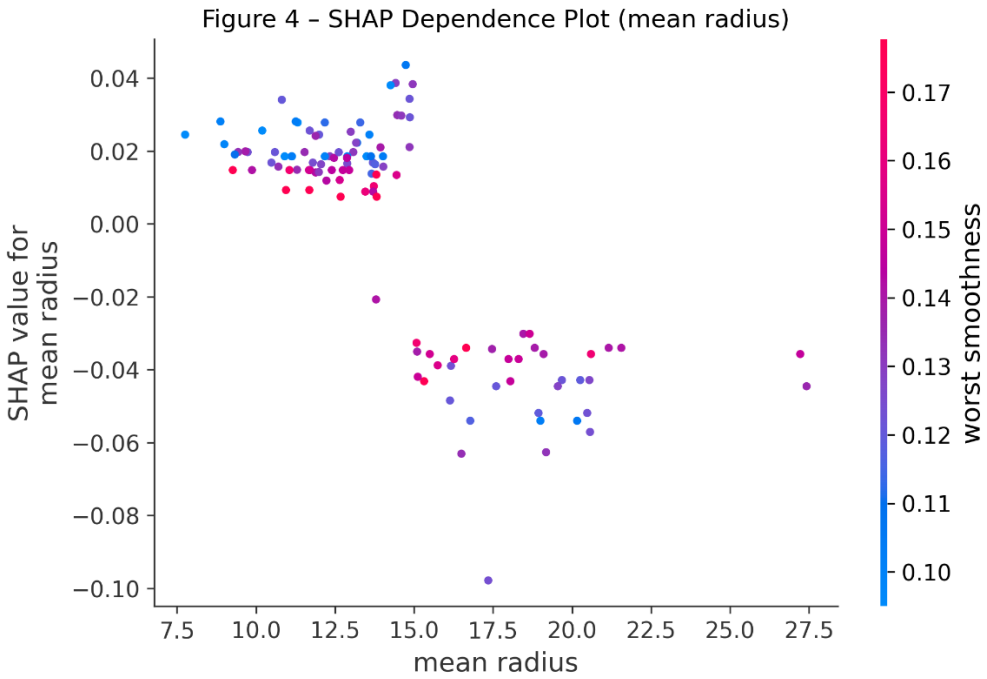
Figure 3 – SHAP Beeswarm Plot:



The Beeswarm plot illustrates the distribution of SHAP values for each feature. Features with a wider spread have a more variable impact on predictions, indicating that their effect changes depending on the data instance.

This highlights the model's ability to capture complex and non-uniform relationships, where the same feature may contribute differently across observations.

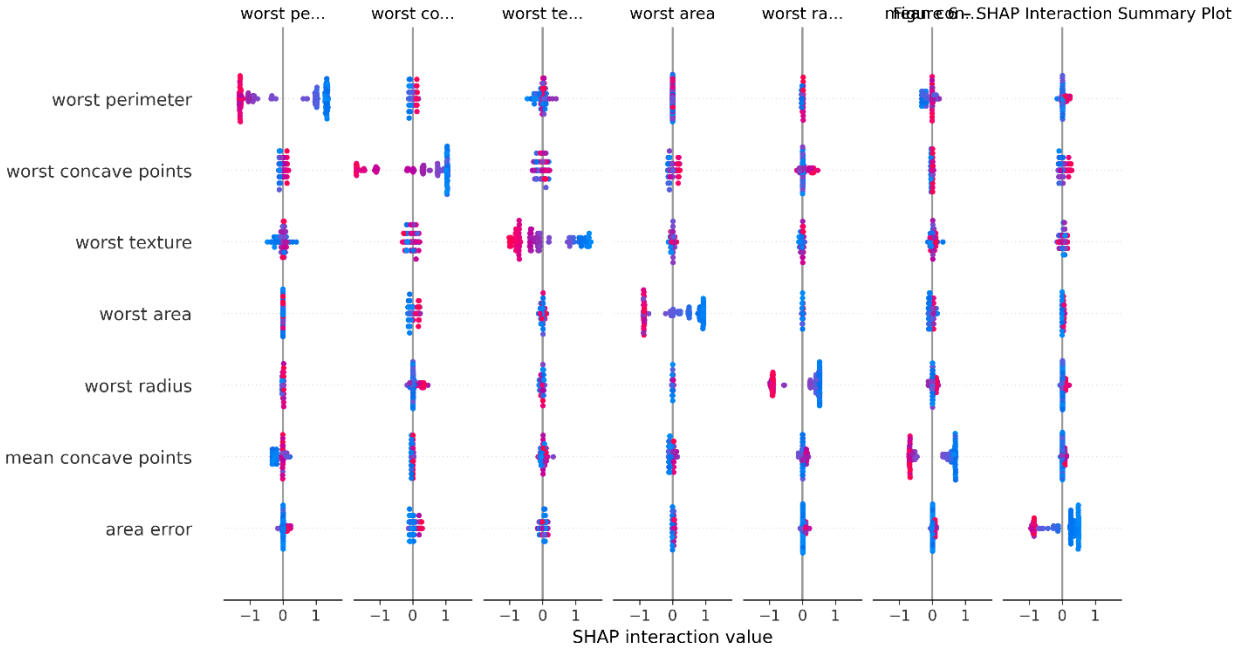
Figure 4 – SHAP Dependence Plot:



The dependence plot shows how the model forecast depends on any given feature.

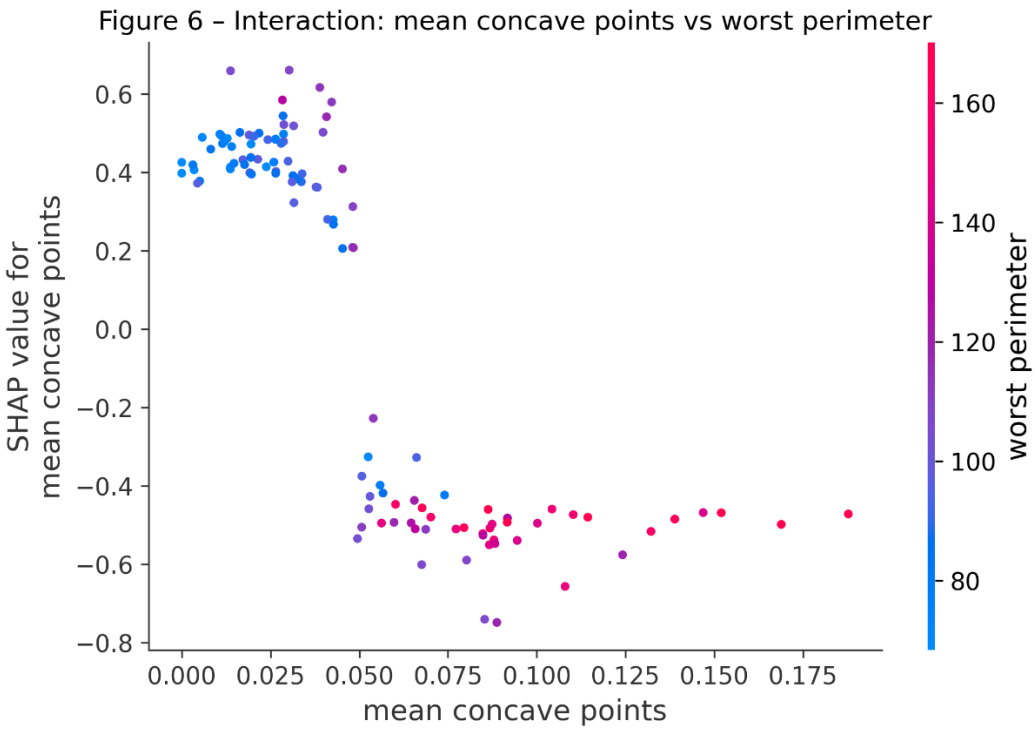
The feature has a clear interactive role with the relationship that is non-linear; a change in the value does not mean a constant change in prediction. The variation in colour also suggests that this feature interacts with another one, indicating that the effect of one variable can depend on another.

Figure 5 – SHAP Interaction Summary:



This graph shows how the pairs of features interact with each other. It shows that certain features jointly impact predictions. The model picks up interactions between features as opposed to independence. This gives more understanding into the inner workings of the model.

Figure 6 – Feature Interaction Plot:



This plot demonstrates the strongest interaction detected in the model. It shows how the effect of one feature depends on the value of a second feature, thereby identifying hidden relationships which are not usually seen otherwise. This reveals that feature interactions are quite important for what the model is doing.

5. Discussion

The findings indicate that SHAP gives far better insight into model behaviour than standard feature importance techniques do.

While the feature importance plot of XGBoost shows important variables such as mean radius and mean texture, it does not explain the impact of these features. On the other hand, SHAP provides the direction and magnitude of each contribution, bringing us closer to the truth.

One key takeaway from this analysis is the existence of non-linear relationships. The dependence plots confirm that a change in the feature led to an uneven change in the prediction. This confirms that the model is picking up complex patterns. Tree-based models are better than simpler linear ones, shows this.

In addition, the interaction analysis shows that features do not work in isolation. To specify a multiple covariate formalization in a scientific model because it is never there by itself as other multiple covariates will be there working jointly at the actual scenario like in the case of a medical diagnosis where a symptom does not operate on its own.

From the practical point of view, SHAP improves trust and transparency in M L systems. The tool enables users to confirm the model's decisions and potential biases on a global and local scale. In high-stakes working environments, interpretability matters even more.

The combination of XGBoost and SHAP can, therefore, be quite useful in achieving a more balanced feature-performance marginal. They can also help with interpretability issues. In summary, they sort increases value in real-world deployment scenarios.

6. Conclusion

- SHAP provides a powerful framework for interpreting complex models.
- It improves transparency and trust in machine learning systems.
- It reveals both global and local feature contributions.
- Interaction analysis uncovers hidden relationships between features.
- Compared to traditional methods, SHAP offers significantly deeper insights

7. Future Work

- Apply SHAP to deep learning models.
- Use real-world clinical datasets.
- Combine SHAP with fairness analysis.
- Develop simplified visual explanations for non-experts.

8. References

Lundberg, S.M. & Lee, S.I. (2017). A Unified Approach to Interpreting Model Predictions. NeurIPS.

Chen, T. & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. KDD.

Scikit-learn Dataset: Breast Cancer Wisconsin Dataset

SHAP Documentation: https://shap.readthedocs.io